POWER BI PROJECT REPORT

Superstore Sales Analysis Dashboard

Business Intelligence • Data Visualization • Performance Analysis

Project type	Sales and profitability analysis using Microsoft Power BI
Dataset	superstore_sales_clean(1).csv
Reporting period	2015–2018 (reported coverage; verify after loading)
Dashboard pages	Executive Overview • Regional Performance • Product Performance

Prepared as project documentation for an internship task. This report describes the proposed Power BI model, measures, dashboard design, and validation approach. Confirm all KPI values against the final refreshed PBIX file.

Report date: 25 October 2026

1. Executive Summary

This project uses Microsoft Power BI to transform a Superstore sales dataset into an interactive report for monitoring revenue, profitability, order activity, customer reach, and product performance. KPI cards, charts, tables, and slicers help users move from overall performance to regional and product-level detail.

The proposed report has three pages: Executive Overview for a high-level summary, Regional Performance for geographic comparisons, and Product Performance for category, sub-category, and individual product analysis. Shared slicers can keep exploration consistent across pages.

Project objectives

• Summarize sales, profit, orders, customers, quantity, and profit margin.

• Compare results across regions, states, categories, and sub-categories.

• Identify profitable products and products with low or negative profit.

• Provide interactive filtering to support business analysis.

• Validate figures and field types before presenting conclusions.

2. Dataset and Preparation

The source file is identified as **superstore_sales_clean(1).csv**. Referenced columns include Order ID, Order Date, Ship Date, Customer ID, Customer Name, Region, State, City, Segment, Category, Sub-Category, Product Name, Sales, Profit, Quantity, Discount, and Ship Mode. Verify exact column names in Power BI.

Field group	Example columns	Use
Identifiers	Order ID, Customer ID	Distinct order/customer counts
Dates	Order Date, Ship Date	Time trends and date filters
Geography	Region, State, City	Regional comparisons and maps
Product	Category, Sub-Category, Product Name	Product mix and profitability
Measures	Sales, Profit, Quantity, Discount	KPIs and charts

Data preparation checklist

• Import via Home → Get data → Text/CSV.
• Verify Sales, Profit, Quantity, and Discount are numeric.
• Set Order Date and Ship Date to Date.
• Check blanks, duplicates, and inconsistent category labels.
• Treat Order ID and Customer ID as identifiers, not quantities to sum.
• Use the actual imported table name in DAX. FactOrders is only an optional example name.

3. Core DAX Measures

Create each measure through Modeling → New measure. Replace **YourTable** with the actual table name displayed in the Data pane.

Total Sales

```
Total Sales = SUM('YourTable'[Sales])
```

Total Profit

```
Total Profit = SUM('YourTable'[Profit])
```

Total Orders

```
Total Orders = DISTINCTCOUNT('YourTable'[Order ID])
```

Total Customers

```
Total Customers = DISTINCTCOUNT('YourTable'[Customer ID])
```

Total Quantity

```
Total Quantity = SUM('YourTable'[Quantity])
```

Profit Margin %

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

Formatting: set Sales and Profit to currency, counts to whole numbers, and Profit Margin % to percentage with one or two decimals. Margin should be total profit divided by total sales, not an average of row-level margins.

Modeling note

A single imported fact table is sufficient for this beginner-friendly report. Separate date, customer, product, or geography dimensions are optional enhancements, not prerequisites.

4. Page 1 — Executive Overview

Purpose: summarize business performance and show how sales change over time.

Visual	Fields / setup	Question answered
KPI cards	Total Sales, Total Profit, Total Orders, Total Customers, Total Quantity, Profit Margin %	What is the overall performance?
Line chart	Order Date by month; Total Sales as values	How do sales change over time?
Bar/column chart	Category by Total Sales	Which categories contribute revenue?
Bar/column chart	Region by Total Sales	Which regions contribute revenue?
Slicers	Order Date, Region, Segment	How do results change with filters?

Design: use a 16:9 canvas, burgundy title band, readable KPI cards, and a consistent color palette. Keep slicers near the top or along one side.

5. Page 2 — Regional Performance

Purpose: compare sales and profit across regions and states.

Visual	Fields / setup	Business use
KPI cards	Total Sales, Total Profit, Profit Margin %	Summarize selected geography
Clustered bar	Region on Y-axis; Total Sales on X-axis	Compare regional revenue
Clustered bar	Region on Y-axis; Total Profit on X-axis	Compare regional profit
Map (optional)	State as Location; Total Sales as Size; Profit and margin in tooltips	Explore geography
Matrix	Rows: Region, State; values: Sales, Profit, Margin, Orders	Inspect state results
Slicers	Order Date, Region, Segment	Filter results

If maps are unavailable, use a State by Total Sales bar chart and apply a Top N = 10 filter. Set the State field's geographic data category if required.

6. Page 3 — Product Performance

Purpose: compare categories and sub-categories, inspect products, and find low- or negative-profit items.

Visual	Fields / setup	Business use
KPI cards	Total Sales, Total Profit, Total Quantity, Profit Margin %	Summarize product performance
Clustered bar	Category by Total Sales	Compare category revenue
Clustered bar	Sub-Category by Total Profit; sort descending	Find strongest/weakest sub-categories
Scatter chart	X: Total Sales; Y: Total Profit; group by Sub-Category	Spot high-sales, low-profit areas
Table	Product Name, Sales, Profit, Quantity, Profit Margin %	Inspect individual products
Slicers	Order Date, Category, Region	Filter product results

For the scatter chart, group points by Sub-Category or Product Name where supported. If it displays one point only, configure the grouping field or use the profit-by-sub-category bar chart instead.

7. Interactivity and User Experience

• Sync relevant slicers across pages using View → Sync slicers.
• Use descriptive chart titles and axis labels.
• Use tooltips for secondary metrics instead of overcrowding charts.
• Sort comparison charts by the measure being analyzed.
• Keep currency and percentage formats consistent.
• Test cross-filtering by selecting a category or region and checking the other visuals.

8. Validation and Quality Checks

The following values were reported during earlier project guidance. They are reference checks only and must be verified against the imported file and final report.

Metric	Reported reference	How to validate
Total Sales	$2,296,918.84	Compare unfiltered Sales card
Total Profit	$286,409.63	Compare unfiltered Profit card
Total Orders	5,009	Distinct Order ID count
Total Customers	793	Distinct Customer ID count
Total Quantity	37,871	Sum Quantity
Profit Margin	About 12.47%	Profit divided by Sales
Date coverage	2015–2018	Check min/max Order Date

If your values differ, verify the source file, data types, full row load, and active filters. Do not change the data just to force a match.

9. Expected Outcomes

The completed report should help users monitor sales and profit, compare geographic results, understand sales trends, assess category and sub-category performance, and inspect individual products for potential profitability issues.

10. Conclusion

This project demonstrates a practical BI workflow: prepare data, create reusable DAX measures, build interactive visuals, and validate results. The three-page dashboard progresses from executive-level monitoring to regional and product-level analysis. Final conclusions should be based on the refreshed and verified Power BI report.

Recommended build order

1. Import the CSV and verify data types.
2. Create the six core measures.
3. Build Executive Overview.
4. Build Regional Performance.
5. Build Product Performance.
6. Synchronize slicers and format visuals.
7. Validate KPI values and save the PBIX file.

End of report